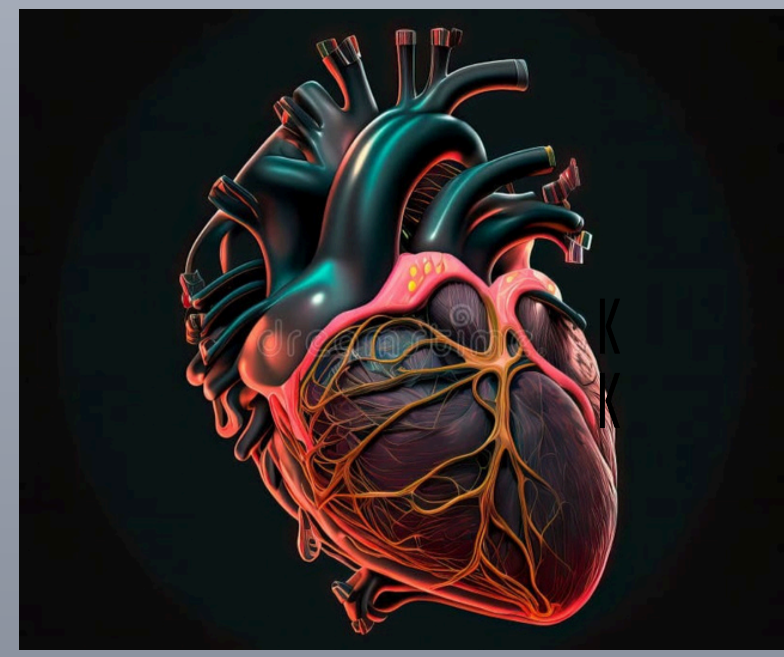
PREDICTING HEART DISEASE

"Detecting the silent killer before it strikes — a heartbeat saved is a life secured."



INTRODUCTION

Heart disease is one of the world's deadliest yet most preventable conditions, often showing no clear symptoms until it's too late. This project leverages machine learning and Python to analyze key health parameters like age, cholesterol, and blood pressure, predicting the risk of heart disease with high accuracy. A user-friendly web app makes these predictions accessible in real time, supporting early diagnosis and timely medical care.

By combining data and technology, we aim to save lives — one prediction at a time.

OBJECTIVE

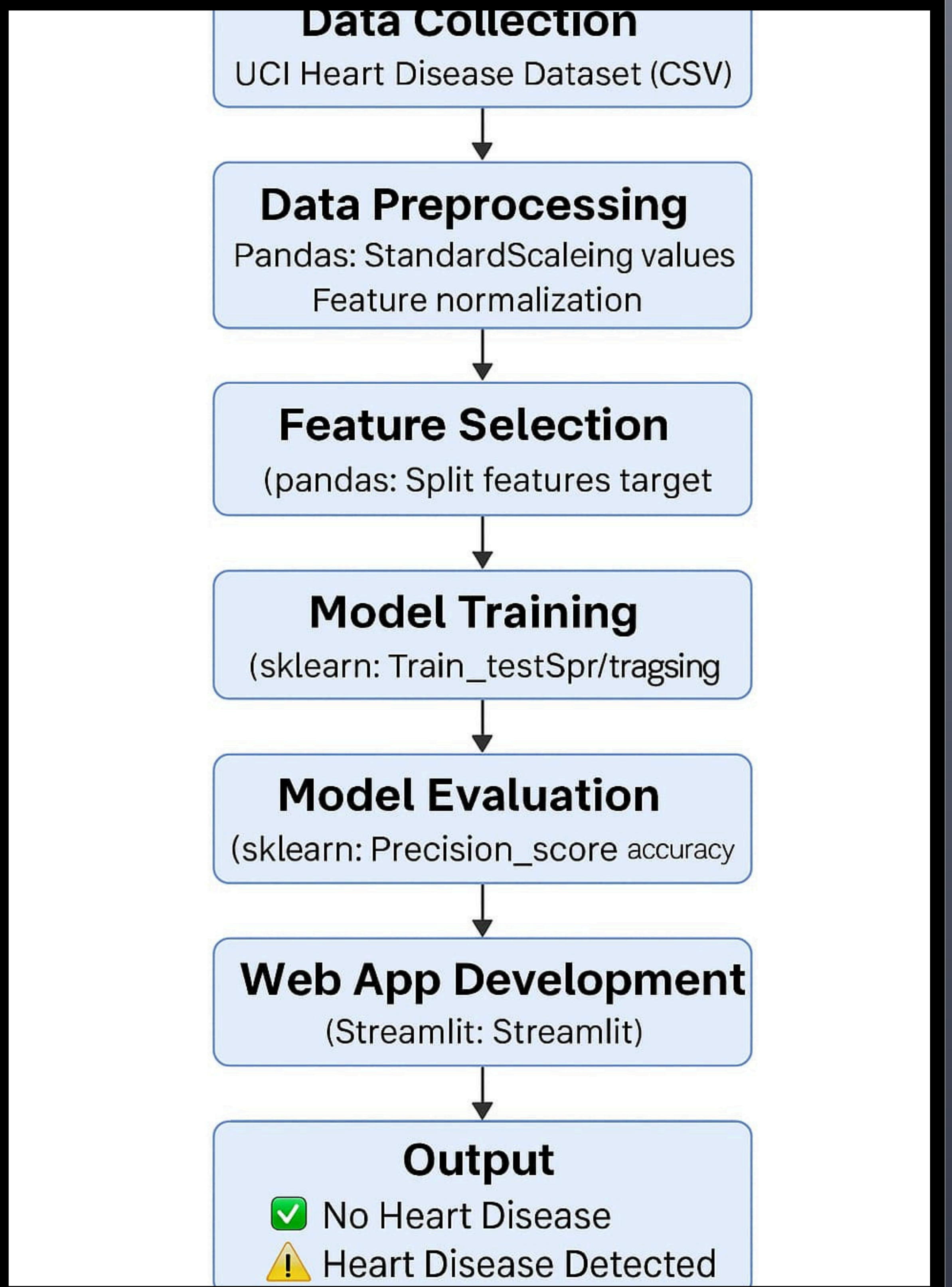
To design a heart disease prediction system using Python and machine learning algorithms that analyze health parameters such as age, blood pressure, and cholesterol.

This project aims to support early diagnosis, improve preventive care, and deliver real-time results through an interactive Python-based web application.

APPLICATION

This project helps in early detection of heart disease, supports remote health checkups, and can be used in health camps, mobile apps, and preventive care systems.

FLOWCHART



CODE

```python
# heart_disease_app.py
import streamlit as st
import pandas as pd
from sklearn.linear_model import LogisticRegression
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split
# Load dataset
@st.cache_data
def load_data():
    df = pd.read_csv("heart.csv")
    return df
df = load_data()
# Train model
X = df.drop("target", axis=1)
y = df["target"]
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2, random_st
model = LogisticRegression()
model.fit(X_train, y_train)
# Streamlit App
st.title("💚 Heart Disease Prediction App")
st.write("Enter the following details:")
# User inputs
age = st.number_input("Age", min_value=1, max_value=120, value=45)
sex = st.selectbox("Sex", ["Male", "Female"])
cp = st.selectbox("Chest Pain Type (cp)", [0, 1, 2, 3])
```

```python
cp = st.selectbox("Chest Pain Type (cp)", [0, 1, 2, 3])
trestbps = st.number_input("Resting Blood Pressure (trestbps)", value=120)
chol = st.number_input("Serum Cholesterol in mg/dl (chol)", value=200)
fbs = st.selectbox("Fasting Blood Sugar > 120 mg/dl (fbs)", [0, 1])
restecg = st.selectbox("Resting ECG Results (restecg)", [0, 1, 2])
thalach = st.number_input("Max Heart Rate Achieved (thalach)", value=150)
exang = st.selectbox("Exercise Induced Angina (exang)", [0, 1])
oldpeak = st.number_input("Oldpeak (ST depression)", value=1.0)
slope = st.selectbox("Slope of ST segment (slope)", [0, 1, 2])
ca = st.selectbox("Number of major vessels (ca)", [0, 1, 2, 3])
thal = st.selectbox("Thalassemia (thal)", [0, 1, 2, 3])
# Prepare input for model
user_data = pd.DataFrame([[age, 1 if sex == "Male" else 0, cp, trestbps, chol,
                           fbs, restecg, thalach, exang, oldpeak, slope, ca, thal]],
                          columns=X.columns)
user_data_scaled = scaler.transform(user_data)
# Predict
if st.button("Predict"):
    prediction = model.predict(user_data_scaled)[0]
    if prediction == 1:
        st.error("⚠ The model predicts **Heart Disease**.")
    else:
        st.success("☑ The model predicts **No Heart Disease**.")
st.markdown("---")
st.markdown("Made with ❤   using Streamlit and Scikit-learn")
```

RESULT